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(54) **MAGNETICALLY COUPLED
GALVANICALLY ISOLATED
COMMUNICATION USING LEAD FRAME**

(58) **Field of Classification Search**
None
See application file for complete search history.

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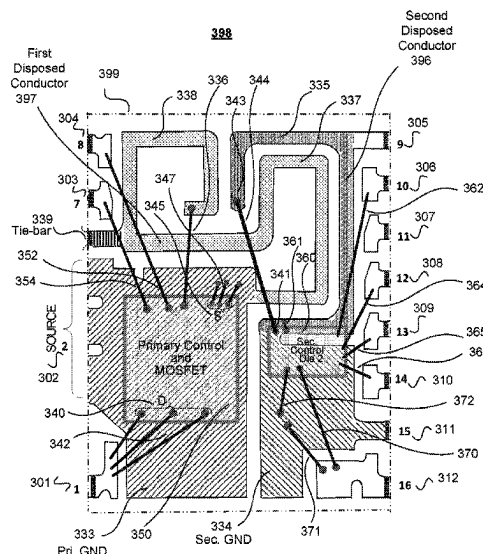
(57) **ABSTRACT**

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An integrated circuit package includes a lead frame and an encapsulation that substantially encloses the lead frame. The lead frame further includes a first conductor comprising a first conductive loop and a second conductor galvanically isolated from the first conductor, proximate to and magnetically coupled to the first conductive loop to provide a communication link between the first and second conductor. The second conductor includes a first conductive portion, a second conductive portion, and a wire coupling together the first conductive portion and the second conductive portion.

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12 Claims, 7 Drawing Sheets



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continuation of application No. 16/431,156, filed on Jun. 4, 2019, now Pat. No. 10,833,590, which is a continuation of application No. 15/943,465, filed on Apr. 2, 2018, now Pat. No. 10,361,632, which is a continuation of application No. 15/096,515, filed on Apr. 12, 2016, now Pat. No. 10,079,543, which is a continuation of application No. 14/688,257, filed on Apr. 16, 2015, now Pat. No. 9,331,004, which is a continuation of application No. 13/677,088, filed on Nov. 14, 2012, now Pat. No. 9,035,435.

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H01L 25/18 (2023.01)
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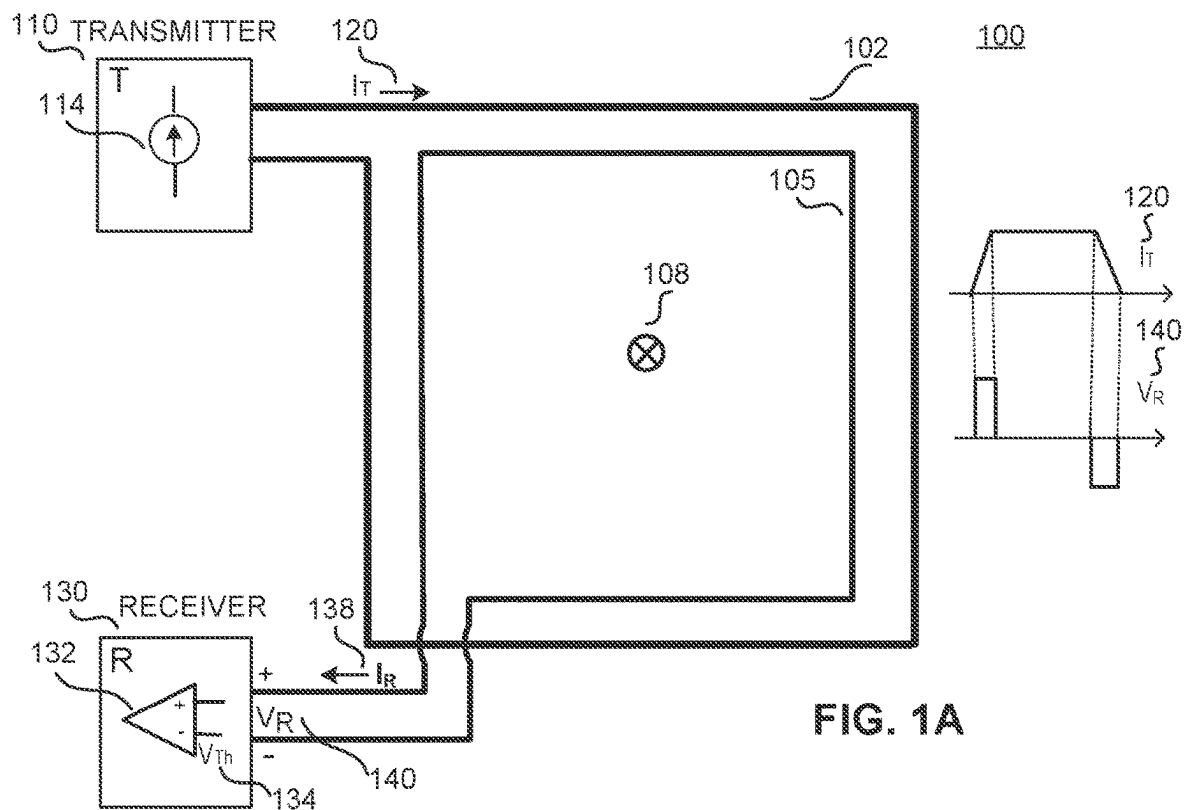


FIG. 1A

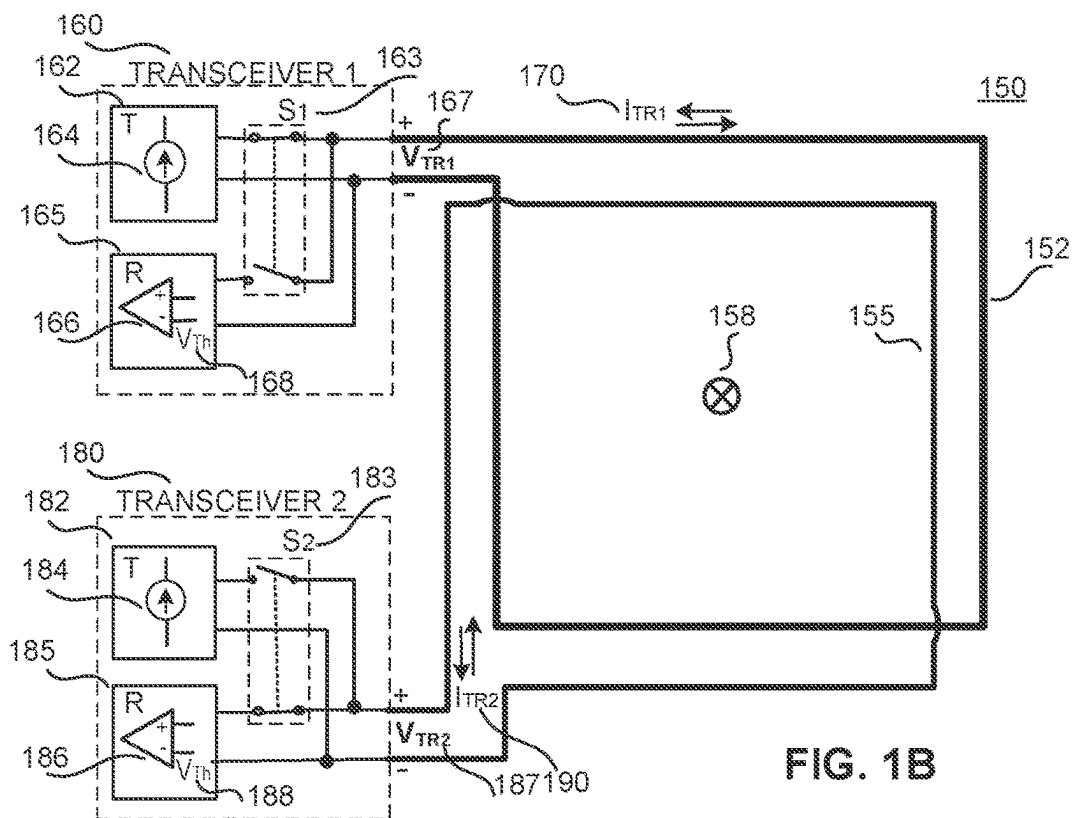


FIG. 1B

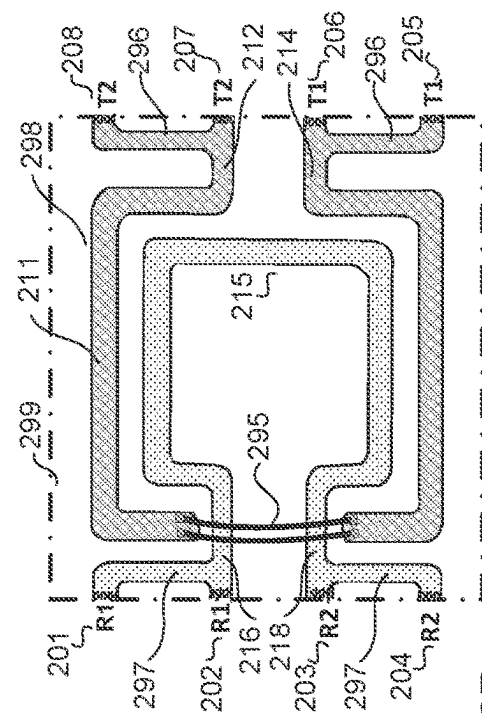


FIG. 2A

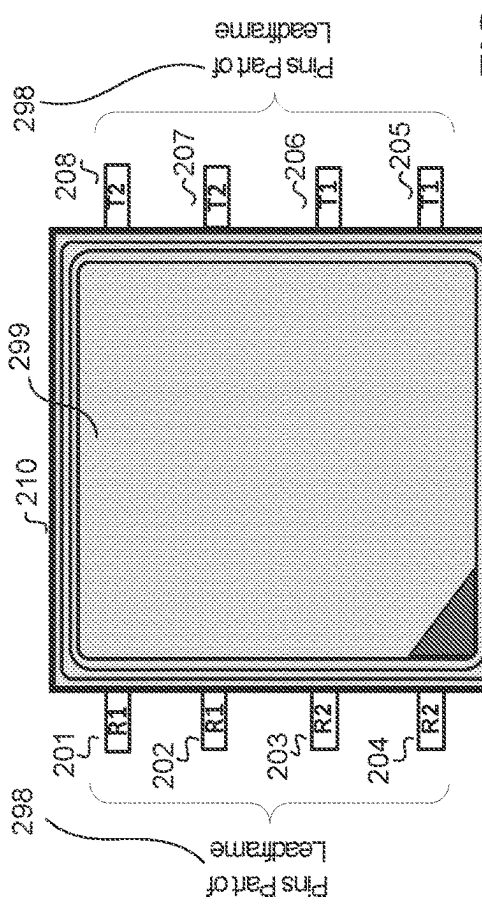


FIG. 2B

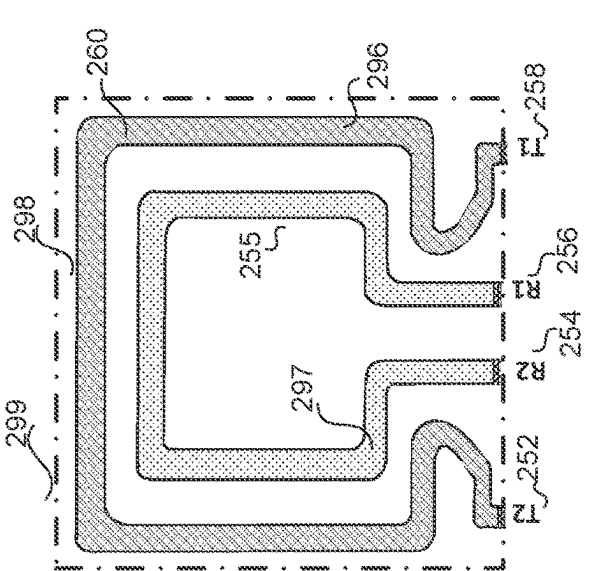


FIG. 2C



FIG. 2D

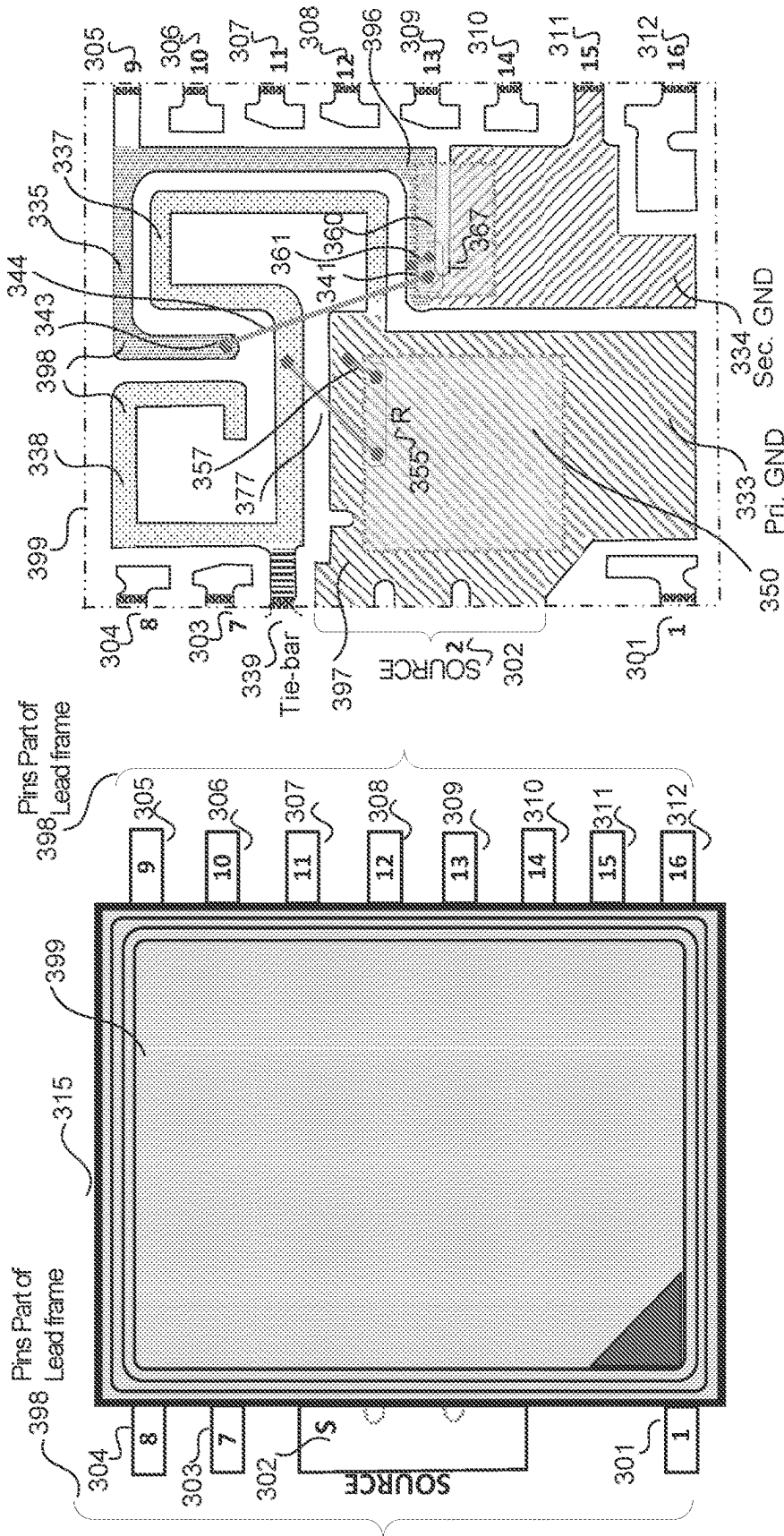


FIG. 3A

FIG. 3B

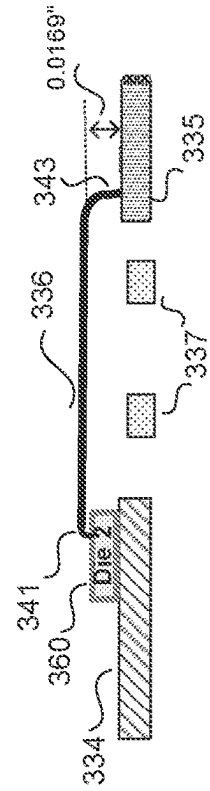


FIG. 3D

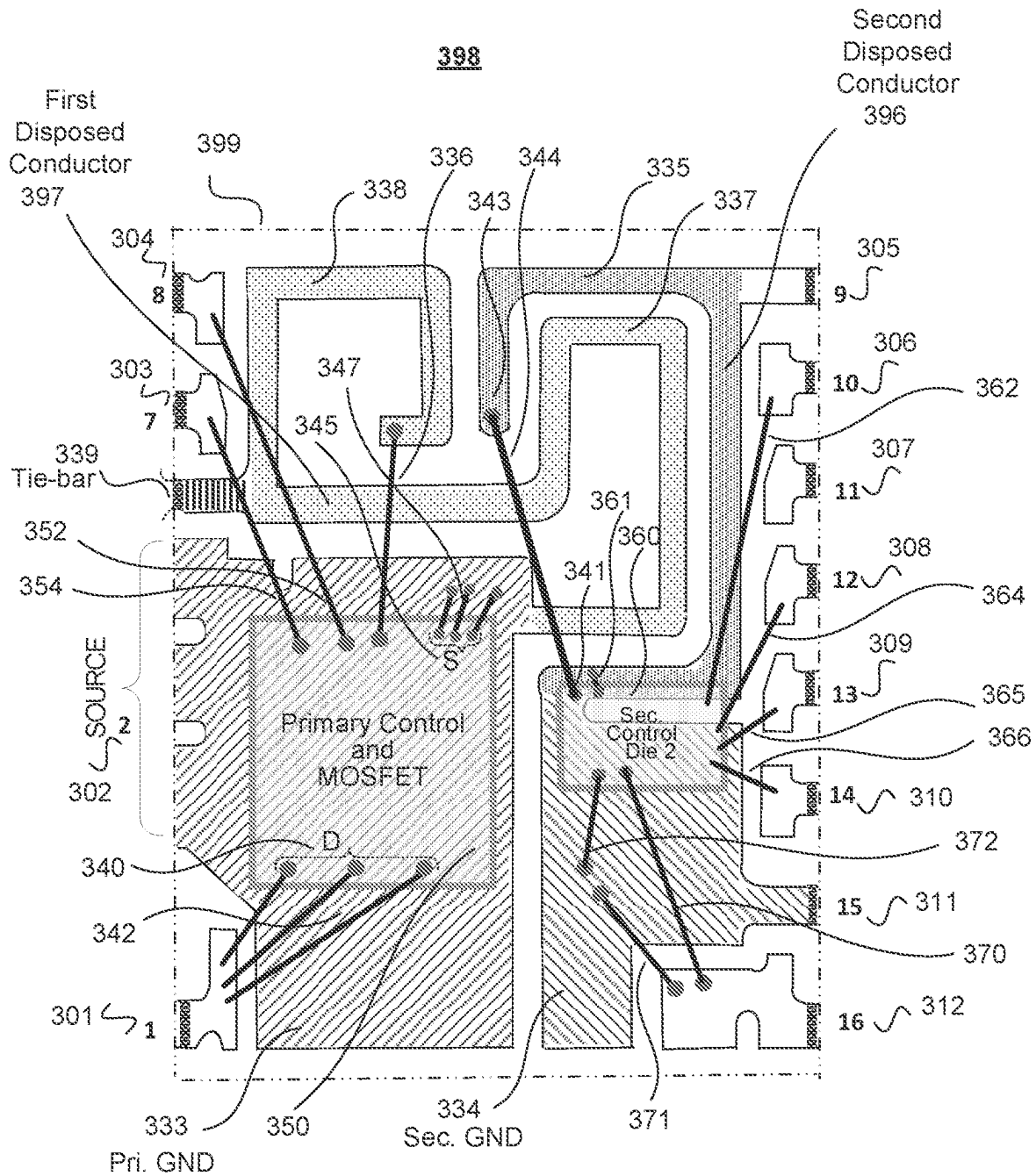


FIG. 3C

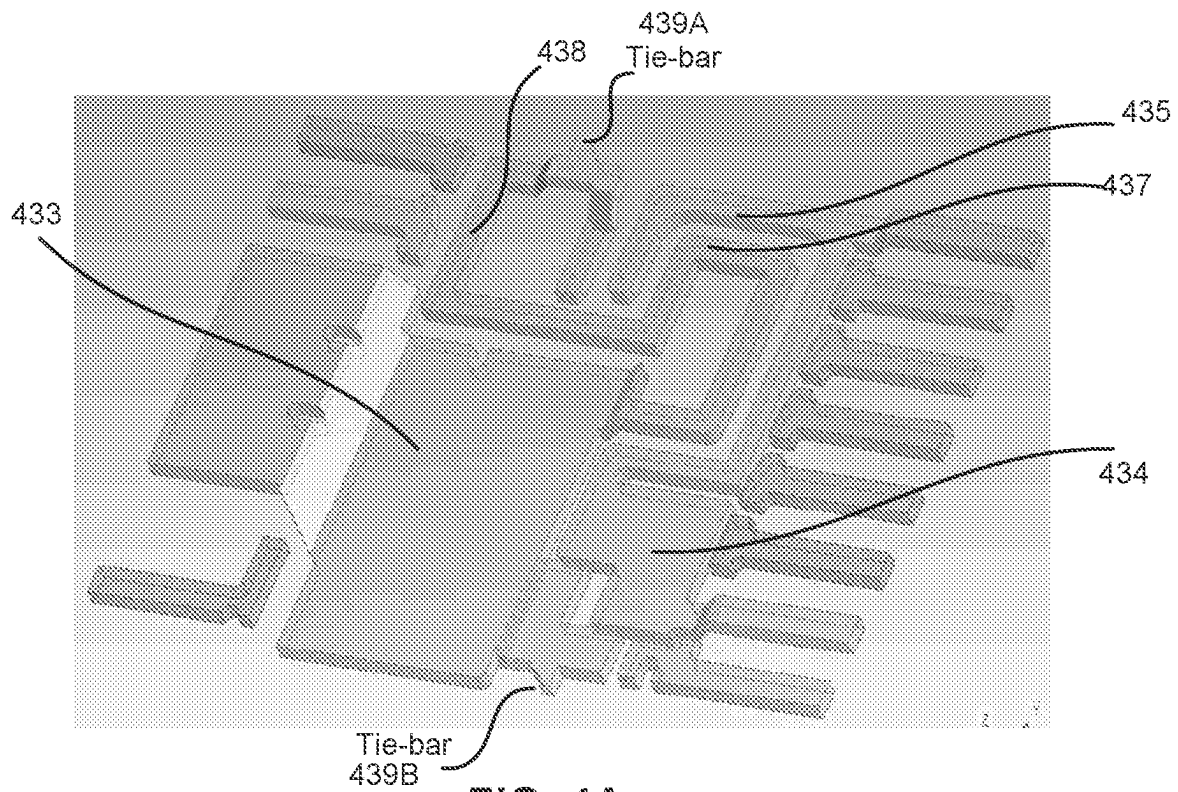


FIG. 4A

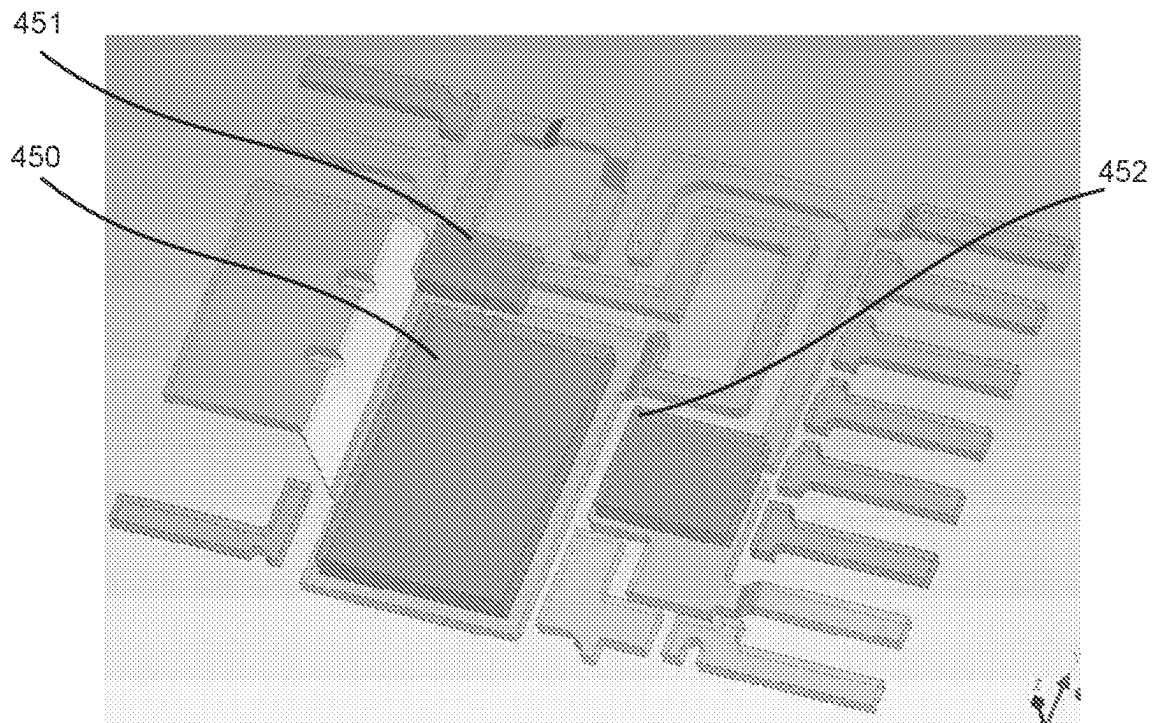
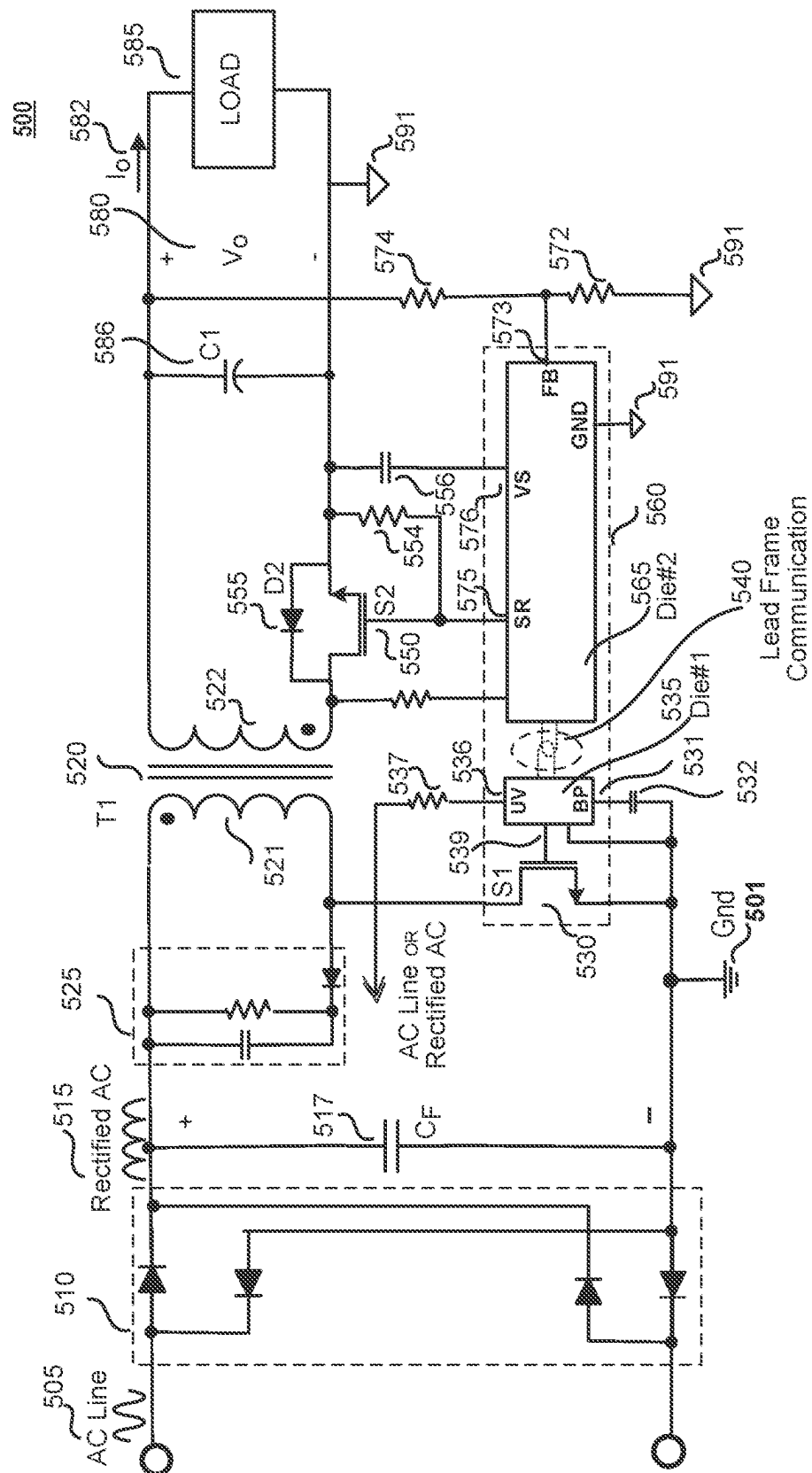
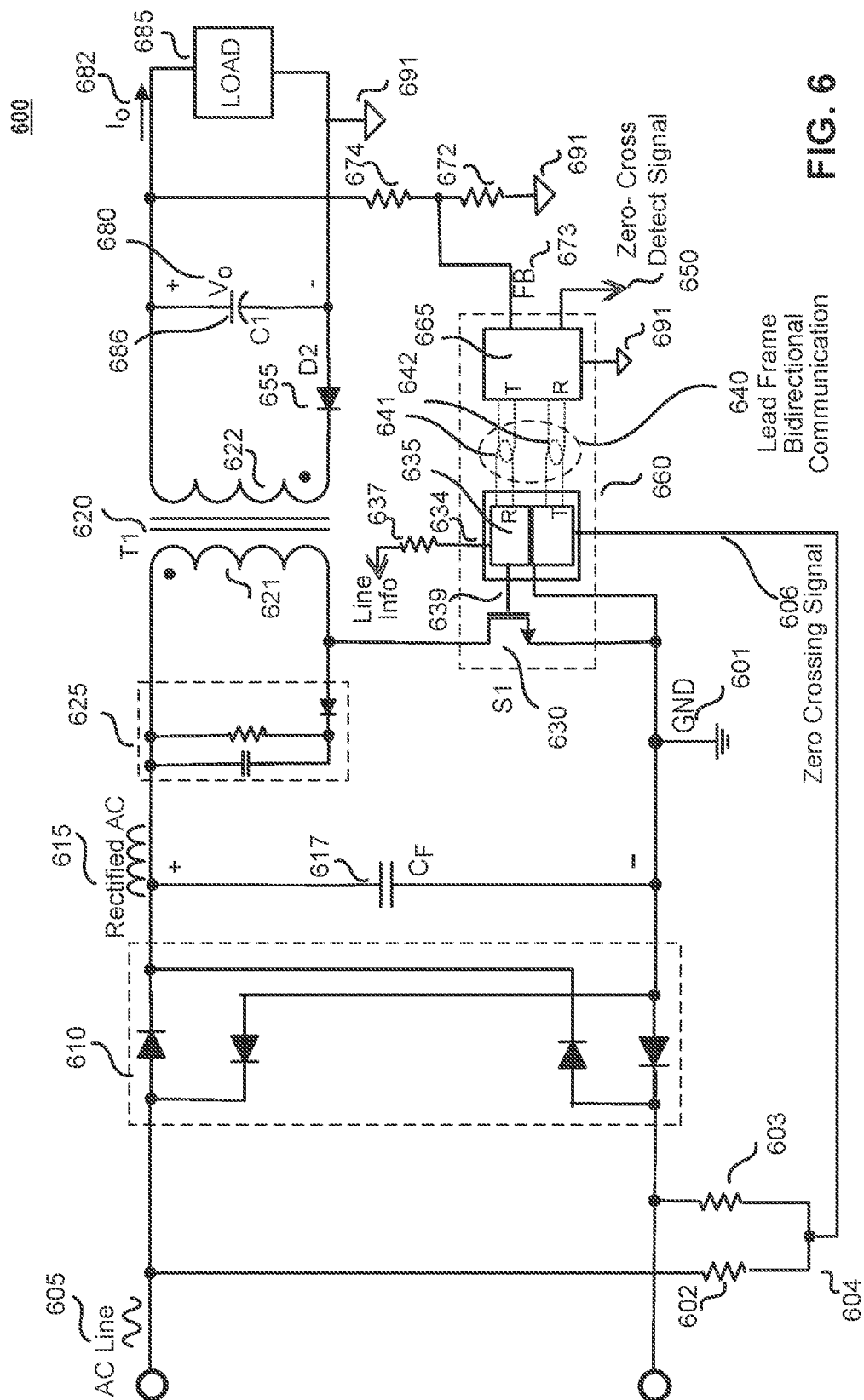


FIG. 4B



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MAGNETICALLY COUPLED GALVANICALLY ISOLATED COMMUNICATION USING LEAD FRAME

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/029,302, filed on Sep. 23, 2020, which is a continuation of U.S. patent application Ser. No. 16/431,156, filed Jun. 4, 2019, now U.S. Pat. No. 10,833,590, which is a continuation of U.S. patent application Ser. No. 15/943,465, filed Apr. 2, 2018, now U.S. Pat. No. 10,361,632, which is a continuation of U.S. patent application Ser. No. 15/096,515, filed Apr. 12, 2016, now U.S. Pat. No. 10,079,543, which is a continuation of U.S. patent application Ser. No. 14/688,257, filed on Apr. 16, 2015, now U.S. Pat. No. 9,331,004, which is a continuation of U.S. patent application Ser. No. 13/677,088, filed on Nov. 14, 2012, now U.S. Pat. No. 9,035,435. U.S. patent application Ser. No. 17/029,302, U.S. Pat. Nos. 10,833,590, 10,361,632, 10,079,543, 9,331,004, and 9,035,435 are hereby incorporated by reference in their entirety.

This application is related to U.S. patent application Ser. No. 13/677,068 of Kung et al., filed Nov. 14, 2012, entitled "Noise Cancellation For A Magnetically Coupled Communication Link Utilizing A Lead Frame," and assigned to the Assignee of the present application.

This application is also related to U.S. patent application Ser. No. 13/677,120 of Balakrishnan et al., filed Nov. 14, 2012, entitled "Switch Mode Power Converters Using Magnetically Coupled Galvanically Isolated Lead Frame Communication," and assigned to the Assignee of the present application.

BACKGROUND INFORMATION

Field of the Disclosure

The present invention relates generally to communication between circuits that require galvanic isolation. More specifically, examples of the present invention are related to communication across an isolation barrier in switch mode power converters such as power supplies and power inverters.

Background

Switch mode power converters are widely used for household or industrial appliances that require a regulated direct current (dc) source for their operation, such as for example battery chargers that are commonly used in electronic mobile devices. Off-line ac-dc converters convert a low frequency (e.g., 50 Hz or 60 Hz) high voltage ac (alternating current) input voltage to a required level of dc output voltage. Various types of switch mode power converters are popular because of their well regulated output, high efficiency, and small size along with their safety and protection features. Popular topologies of switch mode power converters include flyback, forward, boost, buck, half bridge and full bridge, among many others including resonant types.

Safety requirements for isolated switch mode power converters generally require the use of high frequency transformers to provide galvanic isolation between the inputs and outputs of the switch mode power converters in addition to the voltage level change at the output.

A major challenge in the market of switch mode power converters is reducing the size and cost of the switch mode

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power converter while maintaining high performance operating specifications. In known isolated switch mode power converters, the sensing of the outputs of the switch mode power converters and communication of feedback signals for regulating switch mode power converter output parameters such as current or voltage is usually accomplished using external isolation components such as, for example, opto-couplers. These known methods add unwanted additional size as well as cost to switch mode power converters. In addition, opto-couplers are slow in operation and in many cases limit the feedback bandwidth and the transient response of the switch mode power converter.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting and non-exhaustive embodiments of the present invention are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified.

FIG. 1A shows the conceptual operation of magnetically coupled conductive loops transmitting and receiving signals to communicate digital or analog information for the purpose of this disclosure.

FIG. 1B illustrates another conceptual example of conductive loops suitable for a bidirectional operation according to teaching of this disclosure.

FIG. 2A shows an outside view of one example of an integrated circuit package with galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

FIG. 2B shows an inside view of one example structure of galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the example integrated circuit package of FIG. 2A in accordance with the teachings of the present invention.

FIG. 2C shows an outside view of one example of an integrated circuit package with galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

FIG. 2D shows an inside view of one example structure of galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the example integrated circuit package of FIG. 2C in accordance with the teachings of the present invention.

FIG. 3A shows an outside view of an example of an integrated circuit package with a magnetically coupled communication link formed by isolated conductors of the lead frame inside the encapsulated portion the integrated circuit package in accordance with the teachings of the present invention.

FIG. 3B shows an inside view of one example of a lead frame inside an integrated circuit package with a magnetically coupled communication link formed by magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

FIG. 3C shows an inside view of an example of a lead frame of a multi-die isolated controller integrated circuit package with a magnetically coupled communication link between the controller dice, the link being formed by magnetically coupled conductive loops of isolated conduc-

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tors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

FIG. 3D shows an example side-view of a jumper bond wire coupled to an integrated circuit die and a conductive loop of an isolated conductor of a lead frame inside an integrated circuit package in accordance with the teachings of the present invention.

FIG. 4A shows a tilted 3D (3 dimensional) view of an inside view of one example of a lead frame of an integrated circuit package with a magnetically coupled communication link formed by magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

FIG. 4B shows a tilted 3D view of an inside view of one example of a lead frame of a multi-die isolated controller integrated circuit package with a communication link between the controller dice formed by magnetically coupled conductive loops formed by isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

FIG. 5 shows a schematic of one example of a synchronous flyback switch mode power converter with secondary control utilizing one example of a multi-die isolated controller integrated circuit package with a magnetically coupled communication link between the controller dice formed by isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention.

FIG. 6 shows a schematic of one example of a flyback switch mode power converter utilizing one example of a multi-die isolated controller integrated circuit package including a bidirectional magnetically coupled communication link between the controller dice inside the encapsulated portion of the integrated circuit package in which output information is transferred to a primary side through the magnetically coupled communication link and a line zero-cross detection signal is transferred to the secondary side through the magnetically coupled communication link in accordance with the teachings of the present invention.

Corresponding reference characters indicate corresponding components throughout the several views of the drawings. Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present invention.

DETAILED DESCRIPTION

In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, to one having ordinary skill in the art that the specific detail need not be employed to practice the present invention. In other instances, well-known materials or methods have not been described in detail in order to avoid obscuring the present invention.

Reference throughout this specification to “one embodiment”, “an embodiment”, “one example” or “an example”

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means that a particular feature, structure or characteristic described in connection with the embodiment or example is included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodiment”, “in an embodiment”, “one example” or “an example” in various places throughout this specification are not necessarily all referring to the same embodiment or example. Furthermore, the particular features, structures or characteristics may be combined in any suitable combinations and/or subcombinations in one or more embodiments or examples. Particular features, structures or characteristics may be included in an integrated circuit, an electronic circuit, a combinational logic circuit, or other suitable components that provide the described functionality. In addition, it is appreciated that the figures provided herewith are for explanation purposes to persons ordinarily skilled in the art and that the drawings are not necessarily drawn to scale.

In some applications multiple controllers may be housed in a single integrated circuit package. Each controller is fabricated as a semiconductor die. The present application discloses an integrated circuit package structure that enables communication between the controllers with galvanic isolation using magnetic coupling between portions of the lead frame while adding little or no cost to the overall package.

An integrated circuit package typically includes a lead frame. The lead frame provides mechanical support for a single die or for multiple dice that may be housed within the integrated circuit package. In general, the lead frame typically includes a die attach pad to which a semiconductor die may be attached. In addition, the lead frame generally also includes leads that serve as electrical connections to circuits external to the integrated circuit package. The lead frame is generally constructed from a flat sheet of metal. The flat sheet of metal may be stamped, etched, punched, etc., with a pattern, which defines the die attach pads and various leads of the lead frame.

As mentioned above, isolation is often provided in known switch mode power converters using external isolation components such as for example opto-couplers or through the use of an extra bias (e.g., feedback) winding on the transformer core that is magnetically coupled to the secondary winding. These known methods add unwanted additional size as well as overall cost to switch mode power converters. Isolation is provided in examples in accordance with the teachings of the present invention with magnetically coupled conductive loops formed by galvanically isolated conductors of the lead frame inside the encapsulated portion of an integrated circuit package structure, which provides a magnetically coupled communication link between isolated circuits. In various examples, the isolation provided by the magnetically coupled communication link formed by isolated conductors of the lead frame of the integrated circuit package in accordance with the teachings of the present invention may be utilized in a variety of applications including switch mode power converters that require galvanic isolation between the primary and secondary sides of the switch mode power converters. Some example switch mode power converters utilizing an integrated circuit package having a magnetically coupled communication link formed by isolated conductors of the lead frame of the integrated circuit package include, but are not limited to, synchronous flyback, isolated flyback, isolated synchronous flyback, buck, forward, half-bridge and full-bridge topologies in accordance with the teachings of the present invention.

For the purpose of this disclosure, a physical closed path for current is referred to as a loop. A loop may include different elements such as conductors (that in examples of

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this disclosure could be formed by lead frame and bond wires inside an IC package) as well as electrical components that are in path of the circulating current. Each element in the loop forms a part of the loop, and combination of one or more elements in the loop is referred to as a partial loop. In the context of magnetic field coupling, a loop enclosing a magnetic field is typically referred to as having one or more turns. Each turn corresponds to one enclosure of the magnetic field.

FIGS. 1A and 1B show the conceptual operation of magnetically coupled conductive loops transmitting and receiving signals to communicate operational information for example in a controller IC of a switch mode power converter in accordance with the teachings of the present invention. In FIG. 1A the magnetically coupled communication link 100 includes an outer conductive loop 102 coupled to a transmit circuit 110 and an inner conductive loop 105 coupled to a receive circuit 130. The outer conductive loop 102 in one example includes a pulse current source 114, injecting a pulse current 120 to conductive loop 102. In embodiments, the transmit circuit 110 may communicate information utilizing the transmitter current I_T 120. In one example, circuits within transmit circuit 110 may control various properties of the transmitter current I_T 120 to communicate information to the receive circuit 130. When the transmitter current I_T 120 is changing or varying in magnitude over time, it produces a changing magnetic field in the proximity of the conductor of the inner conductive loop 105. Due to the laws of electromagnetic induction, a voltage is generated across a conductor that is subjected to a changing magnetic field. The pulse current I_T 120 in one example has a time when it is rising, a time when it is falling and an amplitude. The changing flux generated by outer conductive loop 102 due to transmitter current I_T 120 has a direction entering the surface of the page. Marker 108 illustrates the overall magnetic field that passes through both transmitter loop 102 and receiver loop 105. In general, the "X" symbol as illustrated for marker 108 denotes magnetic field or flux into the page, while a dot symbol for a marker symbol denotes magnetic field or flux out from the page.

In the embodiment therefore, receiver voltage V_R 140 is induced due to the changing magnetic field generated by changes in current I_T 120 and may result in receiver current I_R 138 in the direction illustrated in FIG. 1A.

The receive circuit 130 may include circuits which may receive the voltage and/or current induced by the transmit circuit 110 and interprets the voltage and/or current as information. Properties of the transmitter current I_T 120 which may be controlled to communicate information may include the magnitude and rate of change of the transmitter current I_T 120. In the example of depicted transmitter current I_T 120 the rising and falling slopes defined by the pulse waveform 120 induce pulsating voltage V_R 140 with a positive amplitude during rising time and a negative amplitude during falling time of the transmitter current pulse waveform 120. The receive circuit 130 in one example may include a comparator 132 responding to a comparison of the amplitude of induced voltage pulses V_R 140 of receive circuit 130 to a threshold voltage V_{Th} 134.

The communicated signals may take the form of digital information or of analog information. In the case of digital information, communication can be in the form of binary signals or more complex encoded digital data as will be known to one skilled in the art. It is appreciated that other communication techniques may be used. In other examples, communication techniques which take advantage of the relationship between the transmitter current I_T 120 and the

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resultant induced receiver voltage V_R 140 and receiver current I_R 138 received by the receive circuit 130 may be utilized.

FIG. 1B illustrates another example of the magnetically coupled communication link 150. In one example communication link 150 could be suited for bidirectional communication and includes two conductive loops. First loop 152 and second loop 155 are positioned to enclose the maximum common magnetic field area. In contrast to the example of FIG. 1A, that could be better suited to a unidirectional communication, loops 152 and 154 of bidirectional example of FIG. 1B have approximately the same dimensions. For the best bidirectional operation, physical symmetry of the loops is important resulting in approximately equal bidirectional behavior. The magnetic field or flux in the first loop 152 and second loop 155 has a direction into the page.

The operational/functional difference between FIG. 1A and FIG. 1B is that in FIG. 1B both first loop 152 and second loop 155 are coupled to the transceiver (transmit/receive) circuits 160 and 180 respectively. Transceiver circuit 1, 160 through the selection switch S1 163 may couple either a transmit circuit 162 or receive circuit 165 to the first loop 152.

Transceiver circuit 2, 180 through the selection switch S2 183 may couple either a transmit circuit 182 or receive circuit 185 to the second loop 155.

If the transceiver circuit 1 160 is coupled as a transmit circuit to inject a current pulse I_{TR1} 170 to the first loop, then transceiver circuit 2 180 through the second loop 155 and switch S2 183 would be coupled as a receive circuit to receive the communicated signal as a current pulse I_{TR2} 190 or as a voltage pulse V_{TR2} 187.

On the other hand if the transceiver circuit 2 180 is coupled as a transmit circuit to inject a current pulse I_{TR2} 190 to the second loop, then transceiver circuit 1 160 through the first loop 152 and by the controlled function of the switch S1 163 would be coupled as a receive circuit to receive the communicated signal as a current pulse I_{TR1} 160 or as a voltage pulse V_{TR1} 167.

The transmit circuits 162 and 182 in the transceiver circuits 160 and 180 could include pulse current sources 164 and 184 respectively and the receive circuits 165 and 185 in the transceiver circuits 160 and 180 could include comparator circuits 166 and 186 with threshold voltages 168 and 188 respectively.

To illustrate an example of practical application in IC industry, FIG. 2A and FIG. 2B show an example integrated circuit package 210 with galvanically isolated magnetically coupled conductive loops formed by galvanically isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention. In example illustrated in FIGS. 2A and 2B, there are external pins 201, 202, 203 and 204, as well as external pins 205, 206, 207 and 208, on two sides of integrated circuit package 210. In the example, all of the external pins are part of the lead frame 298 that comprises the internal conductive elements 296 and 297 that are fundamentally part of integrated circuit package 210 before any bond wires, such as bond wires 295, are introduced, and extend from an encapsulation 299 of integrated circuit package 210 as shown. In one example, lead frame 298 may be comprised of known conductive materials utilized for lead frames in integrated circuit packaging, such as for example copper, and is substantially flat and embedded in a molding compound of integrated circuit package 210. In the example, lead frame 298 provides electrical connectivity to and from circuitry coupled to pins 201 to 208.

of package **210** as well as provides mechanical support for the connection of bond wires **295**.

For the purposes of this disclosure, an “encapsulation” of an integrated circuit package may be considered to be any external body, encasing or molding that surrounds or encloses a portion of the lead frame which may include one or more integrated circuit dice disposed therein, as well as connections from the integrated circuit die pads to the lead frame and pins of the integrated circuit package. An example encapsulation may be made from molded non-ferrous insulating material, plastic, ceramic covers or the like. In some examples, the encapsulation of the integrated circuit package may or may not provide hermetic sealing to protect the items encased therein from external elements.

For the purposes of this disclosure, the term “integrated circuit package” refers to the type of packages used generally for integrated circuits. It is appreciated that some embodiments of this invention may have no integrated circuits in the package such as the examples in FIGS. 2A, 2B, 2C and 2D.

FIG. 2B shows a view inside the encapsulation **299** revealing one example structure of the galvanically isolated magnetically coupled conductive loops **215** and **211** formed by isolated first and second conductors **296** and **297** of lead frame **298** of the example integrated circuit package **210** of FIG. 2A in accordance with the teachings of the present invention. In particular, as shown in the illustrated example, lead frame **298** includes first conductor **297** and a second conductor **296**, which are encapsulated in insulating molding compound material within encapsulation **299**. In one example, first and second conductors **297** and **296** of lead frame **298** may be formed from a flat sheet of metal by etching, stamping, punching, or the like, to form first conductive partial loop **215** in first conductor **297**, and a second conductive partial loop **211** in second conductor **296**. In the example depicted in FIG. 2B, a bond wire **295** is coupled to second conductor **296** as shown to couple together portions of second conductive partial loop **211**. In the depicted example, second conductor **296** is galvanically isolated from first conductor **297**. In one example, bond wire **295** has a sufficient path length to provide sufficient isolation space from first conductor **297** in order to maintain the galvanic isolation between first conductor **297** and second conductor **296**. In another example not illustrated, it is appreciated that one or more additional bond wires may be included coupling together portions of first conductive loop **215** and/or second conductive loop **211**. It is appreciated that circuit elements connected between pins **201**, **202** and pins **203**, **204** are needed to complete the partial conductive loop **215** shown in order to transmit or receive signals through the communication link. Likewise it is appreciated that circuit elements connected between pins **205**, **206** and pins **207**, **208** are needed to truly complete the partial conductive loop **211** shown in order to transmit or receive signals through the communication link. However for the purposes of this description, partial conductive loops **211** and **215** may be referred to as conductive loops. It is appreciated that this comment also extends to the subsequent discussion of FIG. 1C and FIG. 1D below.

As shown in the example, second conductive loop **211** is disposed within encapsulation **299** proximate to and magnetically coupled to a first conductive loop **215** to provide a communication link between the galvanically isolated first conductor **297** and second conductor **296** in accordance with the teachings of the present invention. In one example, magnetically coupled portions of the first and second conductive loops **215** and **211** are substantially flat and disposed

substantially in a same plane. As shown in the illustrated example, the first and second conductive loops **215** and **211** each consist of one turn. In one example, the communication link provided by the magnetic coupling between second conductive loop **211** and first conductive loop **215** is utilized to communicate one or more signals between galvanically isolated second conductor **296** and first conductor **297** of the lead frame **298** in accordance with the teachings of the present invention. In one example, a transmitting signal is applied between first terminal **T1** pins **205** and **206**, and second terminal **T2** pins **207** and **208**. It is sometimes desirable to have more than one external pin common to a terminal for convenience of assembly on a circuit board.

Continuing with the illustrated example, the signal is received by first conductive loop **215** through the magnetic coupling from second conductive loop **211** between first terminal **R1** pins **201** and **202**, and second terminal **R2** pins **203** and **204**. In another example, it is appreciated that the signal can also be communicated in the opposite direction to provide bidirectional communications.

As shown in the example, first terminal **R1** pins **201** and **202** are coupled to first conductive loop **215** through a lead frame connection **216** and second terminal **R2** pins **203** and **204** are coupled to first conductive loop **215** through a lead frame connection **218**. In the example, the signals at terminals **T1** and **R1** are in phase in accordance with magnetic coupling and induction laws. In one example, the connections of each terminal **T1** **214**, **T2** **212**, **R1** **216** and **R2** **218** to the respective pairs of outside pins **205/206**, **207/208**, **201/202** and **203/204**, as described above, by providing multiple assembly options simplifies the physical connections on a circuit board on which integrated circuit package **210** is mounted.

FIG. 2C shows an outside view of one example of an integrated circuit package **250** with galvanically isolated magnetically coupled conductive loops formed by isolated conductors of the lead frame **298** inside the encapsulated portion of the integrated circuit package **250** in accordance with the teachings of the present invention. It is appreciated that integrated circuit package **250** of FIG. 2C shares many similarities with integrated circuit package **210** of FIG. 2A. For instance, integrated circuit package **250** of FIG. 2C includes an encapsulation **299** in which a lead frame **298** is disposed. However, one difference is that instead of having external pins arranged at two sides of the integrated circuit package, integrated circuit package **250** includes external pins **252**, **254**, **256** and **258** arranged on one side of integrated circuit package **250**. In the example, all of the external pins are part of the lead frame **298** of integrated circuit package **250** and extend from a single side of the encapsulation **299** of integrated circuit package **250** as shown.

FIG. 2D shows a view inside the encapsulation **299** of one example structure of galvanically isolated magnetically coupled conductive loops **215** and **211** formed by the isolated first and second conductors **297** and **296** of the lead frame **298** of the example integrated circuit package of FIG. 2C in accordance with the teachings of the present invention. It is appreciated that the view inside the encapsulation **299** of integrated circuit package **250** shares many similarities with the view inside the encapsulation **299** of integrated circuit package **210**. For instance, as shown in FIG. 1D, lead frame **298** includes first conductor **297** and a second conductor **296** encapsulated in insulating material within encapsulation **299**. In the depicted example, second conductor **296** is galvanically isolated from the first conductor **297**. As shown in the example, a second conductive loop **260** of

second conductor 298 is disposed within encapsulation 298 proximate to and magnetically coupled to a first conductive loop 255 included in first conductor 297 to provide a communication link between the galvanically isolated first conductor 297 and second conductor 296 in accordance with the teachings of the present invention. One difference from the example illustrated in FIG. 2B is that in the example illustrated in FIG. 2D, there is no bond wire 295 included in first conductive loop 255 and/or second conductive loop 260.

In the example illustrated in FIG. 2D, the communication link provided by the magnetic coupling between second conductive loop 260 and first conductive loop 255 is utilized to communicate one or more signals between galvanically isolated second conductor 296 and first conductor 297 of the lead frame 298 in accordance with the teachings of the present invention. In the example, the transmitting signal is applied between first terminal T1 pin 258 and second terminal T2 pin 252. As shown in the example, first terminal T1 pin 258 and second terminal T2 252 are coupled to second conductive loop 260. The signal is received by first conductive loop 255 through the magnetic coupling from second conductive loop 260 between first terminal R1 pin 256 and second terminal R2 pin 254. In another example, it is appreciated that the signal can also be communicated in the opposite direction to provide bidirectional communications.

FIG. 3A shows an outside view of an example of an integrated circuit package 315 with a magnetically coupled communication link formed by isolated conductive loops of the lead frame 398 inside the encapsulation 399 of the integrated circuit package 315 in accordance with the teachings of the present invention. In the example illustrated in FIG. 3A, there are external pins 301, 302, 303, 304, 305, 306, 307, 309, 310, 311 and 312 as shown. In the example, all of the external pins are part of the lead frame 398 of integrated circuit package 315 and extend from the encapsulation 399 of integrated circuit package 315 as shown. In one example, lead frame 398 may be comprised of known conductive materials utilized for lead frames in integrated circuit packaging, such as for example copper, and is substantially flat and encapsulated in a molding compound. In the example, lead frame 398 provides electrical connectivity to and from internal circuitry within encapsulated portion of the integrated circuit package 315 as well as provides mechanical support for integrated circuits and bond wires inside package 315.

FIG. 3B shows a view inside the encapsulation 399 revealing one example the structure of galvanically isolated magnetically coupled conductive loops 337 and 335 formed by isolated first and second conductors 397 and 396 of the lead frame 398 of the example multi-die isolated controller integrated circuit package 315 of FIG. 3A in accordance with the teachings of the present invention. In particular, as shown in the illustrated example, lead frame 398 includes first conductor 397 and a second conductor 396 encapsulated in insulating material within encapsulation 399. As shown in the depicted example, a first conductor 397 includes a first conductive loop 337 and second conductor 396 includes a second conductive loop 335. As shown in the example, second conductive loop 335 is disposed within encapsulation 399 proximate to and magnetically coupled to a first conductive loop 337 to provide a communication link between the galvanically isolated first conductor 397 and second conductor 396 in accordance with the teachings of the present invention. In one example, first conductor 397 also includes an optional third conductive loop 338, which

in one example may be utilized for noise cancellation and is attached to tie bar 339 as shown. In one example, tie-bar 339 provides a mechanical support connection during the manufacture of package 315 before the lead frame 398 is encapsulated with encapsulation 399. In one example the encapsulation 399 is injection molded with a molding compound. The communication link provided by the magnetic coupling between second conductive loop 335 and first conductive loop 337 is utilized to communicate one or more signals between the galvanically isolated second conductor 396 and first conductor 397 of the lead frame 398 in accordance with the teachings of the present invention.

In FIG. 3B the current signal from transmit circuit 367 that in one example is included in controller die 360, assembled on the die pad 334, is injected through the bond wire 344 from node 341 of transmit circuit 367. The current signal flows to the end node 343 of the bond wire 344 and then completes the second loop 335, flowing through the lead frame back to the transmit circuit 367 through the bond wire 361. The injected signal generates a changing magnetic field that induces a voltage signal in the first conductive loop 337 and results in a current signal closing from the first conductive loop 338 to the receive circuit 335 through the bond wires 377 and 357. The receive circuit 335 may be included in the first controller die 355 assembled on the die pad 333 that is the primary ground.

FIG. 3C shows another view inside the encapsulation 399 in which a first isolated control die 350 is mounted on and coupled to the first conductor 397 and a second isolated control die 360 is mounted on and coupled to the second conductor 396 in accordance with the teachings of the present invention. In the illustrated example, first isolated control die 350 is mounted on die pad 333 and second isolated control die 360 is mounted on die pad 334 as shown. In the illustrated example, die pads 333 and 334 are utilized as isolated primary and secondary ground pads, respectively. In the example shown in FIG. 3C, a magnetically coupled communication link between the first isolated control die 350 and second isolated control die 360 is formed by the magnetically coupled communication link between the first conductive loop 337 and second conductive loop 335 in accordance with the teachings of the present invention. In one example, multi-die isolated controller integrated circuit package 315 may be utilized in a switch mode power converter such as for example a synchronous flyback switch mode power converter with secondary control in accordance with the teachings of the present invention.

Products and applications that require low output voltages, such as for example 5V and below, in some cases, use synchronous rectification to achieve high efficiency and compact form factor. Synchronous rectification utilizes a MOSFET (metal oxide semiconductor field effect transistor) that is switched to behave like a rectifier, in place of an output rectifier diode, to reduce voltage drop and power loss. The switching action of an output MOSFET rectifier is synchronized with the main power switch with well-controlled gating signals. In one example, first isolated control die 350 includes a primary control circuit and a switch (in one example a MOSFET) for use in the primary side of a synchronous flyback switch mode power converter, and the second isolated control die 369 includes a secondary control circuit for use in the secondary side of the synchronous flyback switch mode power converter. In various examples, the primary control circuit and switch/MOSFET may be implemented with a monolithic or hybrid structure for the first isolated control die 350.

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As shown in the example illustrated in FIG. 3C, the primary switch (or MOSFET) is included in first isolated control die 350. In one example, the drain terminal D 340 of the MOSFET is coupled through bond wires 342 to pin 301. The source terminal S 345 of the MOSFET is coupled through bond wires 347 to the primary ground die pad 333, which is accessible through source pin 302. In the illustrated example, there is a wide clearance (i.e., missing pins), often referred to as creepage distance, between drain pin 301 and source pin 302. In the illustrated example, the wide pad of source pin 302 is internally coupled to a primary ground pad 333, which may also serve as a heat sink. In one example, pins 303 and 304 are coupled to first isolated control die 350 through bond wires 352 and 354, respectively, to connect the first isolated control die 350 to external circuitry such as for example line under voltage (example of UV 536 in FIG. 5) and supply bypass capacitor (example of BP 531 in FIG. 5).

Bond wire 336 couples the third conductive loop 338 to the first isolated control die 350. Due to a changing magnetic field generated by a changing current flowing in second conductive loop 335, a voltage signal is induced in the first conductive loop 337. In the example of FIG. 3C first conductive loop 337 is coupled to the third conductive loop 338 (the optional noise cancellation loop that is an extension of the first conductive loop). The induced voltage signal is coupled through bond wires 336 and 347 to the receive circuit that in one example is included in the primary die 350 on the primary ground die pad 333.

In the example, pin 305 is attached to second conductive loop 335 of second conductor 396 for mechanical support. The signal communicated from second isolated control die 360 is coupled to second conductive loop 335 through bond wires 344 and 361, which complete the second conductive loop 335. As shown in the example, bond wire 344 is a connection coupling second conductive loop 335 at point 343 to second isolated control die 360 at point 341. Pin 312 in one example is coupled through the current sense bond wire 371 to the secondary Ground pad 334 and the sensed voltage drop on bond wire 371 is coupled to second isolated control die 360 through bond wires 370 and 372 and is utilized for a secondary current measurement. In one example, the bond wires 362, 364, 365 and 366 are coupled between second isolated control die 360 and pins 306, 307, 308, 309 and 310, respectively, and are utilized for the input/output of secondary signals. In one example, pin 311 provides access to secondary ground pad 334 as shown.

In one example, the slot on secondary ground pad 334 under the second isolated control die 360 makes the second conductive loop 335 longer by forcing the current through the second conductive loop 335 to ground pad 334 to flow closer and parallel to the first conductive loop 337 to improve magnetic coupling. The smaller first conductive loop 337 proximate to and surrounded by the second conductive loop 335 provides a strong magnetic coupling of first and second conductive loops in accordance with the teachings of the present invention. In one example, lead frame 398 is flat, but in other examples some portions of the lead frame 398 may be up set and/or down set for optimum vertical positioning to accommodate die thickness, optimizing bond wire profiles and to align to tie bars and external pins of the integrated circuit package 315.

FIG. 3D shows an example side-view of a bond wire 336, which as mentioned above is an electrical connection, and is coupled to second isolated control die 360 at point 341 and second conductive loop 335 at point 343 of the second conductive loop 335 in accordance with the teachings of the present invention. As shown in the example, bond wire 336

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is at a higher level than the level of second conductive loop 335 and pin pad 334 of lead frame 398. As shown, bond wire 336 has sufficient span to complete second conductive transmitter loop 335 and to be isolated from the first conductive loop 337.

FIG. 4A shows a tilted 3D (3 dimensional) view of an inside view of one example of a lead frame of an integrated circuit package with a magnetically coupled communication link that is formed with magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention. FIG. 4A shows the illustrated lead frame structure shares similarities with the lead frame 398 structures of FIG. 3B and FIG. 3C. In particular, in the example of FIG. 4A, the lead frame structure includes a first conductor including a primary die pad 433 and a first conductive loop 437, as well as a third conductive loop 438, which correspond to die pad 333, first conductive loop 337, as well as third conductive loop 338, respectively, of FIGS. 3B and 3C. In addition, in the example of FIG. 4A, the lead frame structure also includes a second conductor including a secondary die pad 434 and a second conductive loop 435, which correspond to die pad 334 and second conductive loop 335, respectively, of FIGS. 3B and 3C. In FIGS. 4A and 4B the tie-bar connection 439A to support the third conductive loop 438 is at a different location than the tie-bar connection 339 to support the third conductive loop 338 in FIGS. 3B and 3C and the tie-bar connection 439B of FIGS. 4A and 4B is not present in the lead frame 398 of FIGS. 3B and 3C. Consequently, the lead frame design shown in FIGS. 3A, 3B and 3C has no tie bar connections on the top and bottom sides of the encapsulation increasing the external creepage distance between the primary and secondary conductors of the lead frame to the shortest distance between external pins 304 and 305 or between external pins 301 and 312, whichever is smaller, measured along the external surface of the encapsulation.

FIG. 4B shows a tilted 3D view of another inside the encapsulation view of one example of a lead frame of a multi-die isolated controller integrated circuit package with a communication link between the controller dice that are formed with magnetically coupled conductive loops of isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention. FIG. 4B shows a lead frame structure sharing similarities with the lead frame structure shown in FIG. 4A. In the example illustrated in FIG. 4B, the primary and secondary dice are shown mounted on the lead frame. As shown in the illustrated example, the primary switch 450 and controller 451 are on different dice—which is commonly referred to as a hybrid structure. In the example of FIG. 4B, the power MOSFET has a high power rating, which results in a separate die 450 having a large size that covers substantially all of the primary ground die pad 433. In the illustrated example, the primary control die 451 is mounted over part of the third conductive loop 438 as shown. The secondary control die 452 is mounted on the secondary ground die pad 434 as shown.

It is appreciated that an integrated circuit package having a magnetically coupled communication link between galvanically isolated conductors of the lead frame inside the encapsulated portion of the integrated circuit package in accordance with the teachings of the present invention may be utilized in a variety of different applications. Although several different switch mode power converter topologies utilizing such an integrated circuit package having a magnetically coupled communication link are described herein,

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it is appreciated that the specific examples described in this disclosure are provided for explanation purposes, and that other applications may utilize a magnetically coupled communication link between galvanically isolated conductors of a lead frame inside the encapsulated portion of an integrated circuit package in accordance with the teachings of the present invention.

To illustrate, FIG. 5 shows one such example application with a schematic of an example synchronous flyback switch mode power converter 500 with secondary control utilizing one example of a multi-die isolated controller integrated circuit package 560 having a magnetically coupled communication link 540 between the controller dice that is formed with galvanically isolated conductors of a lead frame inside the encapsulated portion of the integrated circuit package 560 in accordance with the teachings of the present invention.

It is appreciated that secondary control for a flyback converter has advantages of tighter output regulation and faster response to load transients. However, as discussed previously, conventional methods of secondary control often use external isolation devices, such as for example opto-couplers, which increase the complexity and cost of the switch mode power converter. By using an example multi-die isolated controller integrated circuit package 560 having a magnetically coupled communication link 540 with isolated primary and secondary control dice, externally added isolation components such as opto-couplers are no longer needed in accordance with the teachings of the present invention. Furthermore, since integrated circuit package 560 provides a magnetically coupled communication link by using the lead frame of the integrated circuit package as discussed previously, galvanic isolation is maintained between the primary and secondary sides of the switch mode power converter at nearly zero additional cost, without having to add external isolation components in accordance with the teachings of the present invention.

In the example synchronous flyback switch mode power converter 500, the primary and secondary controllers are galvanically isolated from one another, but there is still reliable communication between the primary and secondary controllers. It is appreciated that although the example of FIG. 5 shows a synchronous flyback converter, a standard flyback converter, where synchronous MOSFET 550 is replaced by a diode, would also benefit from the teachings of the present invention.

In the example illustrated in FIG. 5, synchronous flyback switch mode power converter 500 includes an input coupled to an ac line 505 as shown. A full-bridge rectifier 510 is coupled to ac line 505 to generate rectified ac 515, which is filtered by capacitance C_F 517. The rectified ac 515 is coupled to be received by energy transfer element 520, which includes a primary winding 521 and a secondary winding 522 as shown. In the illustrated example, clamp circuit 525 is coupled across primary winding 521 of energy transfer element 520 as shown.

In the depicted example, a switching device S1 530 is coupled to the input of synchronous flyback switch mode power converter 500 at the primary ground 501 and to the energy transfer element 520 at primary winding 521. In the illustrated example, switching device S1 530 may be included in a monolithic or hybrid structure in the integrated circuit package 560. As shown in the depicted example, switching device S1 is controlled by control signal 539 from the primary controller die 535 and regulates the energy transfer through primary winding 521 of transformer 520 to the secondary winding 522 in response to line and load

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changes. Clamp circuit 525, which in the illustrated example is a diode-resistor-capacitor circuit, is coupled to clamp the turn-off spikes that result from the leakage inductance from primary winding 521 across the switching device S1 530.

As shown in the example of FIG. 5, switch S2 550 and anti-parallel diode D2 555 are coupled to secondary winding 522 at the secondary side and serve as a synchronous rectifier of synchronous flyback switch mode power converter 500. In one example, the diode D2 555 is an externally connected Schottky diode. In one example, switch S2 550 is controlled by a signal from the SR pin of the secondary controller die 565. Whenever the voltage at SR terminal 575 rises to a value higher than the gate threshold voltage, the synchronous rectifier provided by switch S2 550 begins conducting current. The secondary ripple is smoothed by output filter capacitance C1 586 and the dc output voltage V_o 580 is applied to load 585 with load current I_o 582. The output voltage V_o 580 is sensed through the resistor divider comprised of resistors 572 and 574, which is coupled to the feedback pin FB 573 of the secondary controller. It is appreciated that in other examples resistors 574 and 572 could be integrated within integrated circuit 560 while still benefiting from the teachings of the present invention.

At startup, primary die 535, which is referenced to the primary ground 501, starts the switching of switch S1 530, which starts the transfer of energy to the secondary side. The bypass pin BP 531 is externally coupled to the bypass capacitor 532. The line under voltage pin UV 536 is externally coupled through resistor 537 to the ac input line, which in another example could be coupled to a rectified ac bus 515. Communication between the primary die 535 and secondary die 565 is through a magnetic coupling provided by a magnetically coupled communication link 540 formed by isolated conductors of the lead frame of the integrated circuit package in accordance with the teachings of the present invention. In various examples, the communication link 540 is implemented using galvanically isolated conductive loops included in the lead frame of the integrated circuit package as described above in accordance with the teachings of the present invention.

FIG. 6 shows a schematic of one example of a flyback switch mode power converter 600 utilizing one example of a multi-die isolated controller integrated circuit package including a bidirectional magnetically coupled communication link between the controller dice inside the encapsulated portion of the integrated circuit package in which output information is transferred to a primary side through the magnetically coupled communication link and a line zero-cross detection signal is transferred to the secondary side through the magnetically coupled communication link in accordance with the teachings of the present invention.

In the example illustrated in FIG. 6, flyback switch mode power converter 600 includes an input coupled to an ac line 605 as shown. A full-bridge rectifier 610 is coupled to ac line 605 to generate rectified ac 615, which is filtered by capacitance C_F 617. The rectified ac 615 is coupled to be received by energy transfer element 620, which includes a primary winding 621 and a secondary winding 622 as shown. In the illustrated example, clamp circuit 625 is coupled across primary winding 621 of energy transfer element 620 as shown.

In the depicted example, a switching device S1 630 is included in an integrated circuit package 660. In one example, the switch die and the primary control die may be structured as monolithic or hybrid dice. In the example, switching device S1 630 is coupled to the input of flyback switch mode power converter 600 at the primary ground 601

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and to the energy transfer element 620 at primary winding 621. As shown in the depicted example, switching device S1 630 is controlled by control signal 639 from the primary controller die 635 and regulates the energy transfer through primary winding 621 of transformer 620 to the secondary winding 622 in response to line and load changes. Clamp circuit 625, which in the illustrated example is a diode-resistor-capacitor circuit, is coupled to clamp the turn-off spikes that result from the leakage inductance from primary winding 621 across the switching device S1 630. In the illustrated example, the secondary rectifier diode D2 655 in flyback only conducts current during an off-time of the primary switch 630.

The secondary ripple is filtered by the output filter capacitance C1 686 and the dc output voltage V_o 680 is applied to the load 685 with load current I_o 682. The output voltage V_o 680 is sensed through resistor divider comprised of resistors 672 and 674, which is coupled to the feedback pin FB 673 of the secondary controller die 665 and is referenced to secondary ground 691 isolated from the primary ground 601. In one example, feedback signal 673 is a data signal that is transmitted through the magnetic coupling of the lead-frame communication loop 641 and received by the primary die 635 in reference to the primary ground 601. In one example, the FB signal 673, transferred by lead frame magnetic coupling of the communication link 640 to the primary side controller die 635, may be either a digital or an analog signal. FB signal is utilized in combination with the input line information received at pin 634 through resistor 637 to generate gate control signal 639 to control the switching of switch S1 630 to regulate the transfer of energy through energy transfer element 620 to the output. In one example, lead frame communication link 640 includes unidirectional communication links 641 and 642 to transmit one or more control signals between dice 665 and 635 in accordance with the teachings of the present invention. In another example, lead frame communication link 640 includes a single bidirectional communication link (as depicted in FIG. 1B) using the same magnetically coupled lead frame loop to transmit one or more control signals in either direction between dice 665 and 635 in accordance with the teachings of the present invention.

In one example, the specific control function of the example flyback switch mode power converter 600 of FIG. 6 also utilizes a zero-cross signal of the ac line that is sensed at ac line input 605 through the shunt connected resistors 602 and 603 at the common point 604 referenced to the primary ground 601 as shown. In the example, zero sense signal 606 is coupled to primary die 635 and referenced to primary ground 601, and is transmitted through the magnetic coupling of the lead-frame communication loop 642 and received by the secondary die 665 with reference to the secondary ground 691, which could be utilized as an isolated remote control signal. For example, the zero-cross signal (a pulse synchronous with the ac input voltage passing through zero at every line cycle) could be utilized as an isolated signal for some electric appliances, such as for example washing machines to sense line frequency or generate timing signals necessary for an efficient load switching in the appliance.

As shown in the illustrated example, the lead frame communication link 640 is bidirectional and includes two unidirectional communication links 641 and 642. Communication link 642 is unidirectional in a reverse direction of lead frame communication link 641. It is appreciated that even though in the illustrated example the individual lead frame communication links are described as unidirectional

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communication links, in another example, a single lead frame communication link can be utilized in a bidirectional implementation (as presented, for example, in FIG. 1B) instead of two unidirectional communication links in accordance with the teachings of the present invention.

Even though the magnetically coupled communication link built in accordance with the teachings of this invention provides galvanic isolation, one skilled in the art would appreciate that the overall system, such as a switch mode power converter, that practices this invention need not necessarily be galvanically isolated between the input and output of the system to benefit from the advantages of this invention. For example, in non-isolated converters, a galvanically isolated communication link in accordance with the teachings of the present invention allows communication between two parts of a switch mode power converter that are referenced to different voltages which could be fixed or changing relative each other over time.

The above description of illustrated examples of the present invention, including what is described in the Abstract, are not intended to be exhaustive or to be limitation to the precise forms disclosed. While specific embodiments of, and examples for, the invention are described herein for illustrative purposes, various equivalent modifications are possible without departing from the broader spirit and scope of the present invention. Indeed, it is appreciated that the specific example voltages, currents, frequencies, power range values, times, etc., are provided for explanation purposes and that other values may also be employed in other embodiments and examples in accordance with the teachings of the present invention.

What is claimed is:

1. A lead frame comprising:

a first conductor comprising an inner conductive loop, and a second conductor galvanically isolated from the first conductor, wherein the second conductor comprises an outer conductive loop disposed proximate to and magnetically coupled to the inner conductive loop to provide a communication link between the first and second conductors,

wherein substantially all of the outer conductive loop is outside of the inner conductive loop;

a first ground pad coupled to the first conductor; and a second ground pad comprising a slot formed by an inner portion and an outer portion coupled to the second conductor such that the outer portion extends the outer conductive loop.

2. The lead frame of claim 1 wherein the magnetically coupled portion of the inner and outer conductive loops are disposed substantially in a same plane.

3. The lead frame of claim 2 wherein the inner and outer conductive loops each consist of one turn.

4. The lead frame of claim 2, wherein the inner conductive loop and the outer conductive loop are physically symmetric.

5. The lead frame of claim 2, comprising:

a tie bar connection; and

the first conductor further comprising a third conductive loop coupled to the tie bar connection and configured for noise cancellation.

6. A lead frame comprising:

a first conductor comprising an inner conductive loop; and a second conductor galvanically isolated from the first conductor, wherein the second conductor comprises an outer conductive loop disposed proximate to and mag-

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netically coupled to the inner conductive loop to provide a communication link between the first and second conductors, and

wherein substantially all of the outer conductive loop is outside of the inner conductive loop. 5

7. The lead frame as in claim 6 wherein the inner conductive loop and the outer conductive loop are concentric.

8. The lead frame of claim 7 comprising:

a first ground pad coupled to the first conductor; and 10
a second ground pad comprising a slot formed by an inner portion and an outer portion coupled to the second conductor such that the outer portion extends the outer conductive loop.

9. The lead frame of claim 8 wherein the magnetically 15
coupled portion of the inner and outer conductive loops are disposed substantially in a same plane.

10. The lead frame of claim 9 wherein the inner and outer conductive loops each consist of one turn.

11. The lead frame of claim 9, wherein the inner conduc- 20
tive loop and the outer conductive loop are physically symmetric.

12. The lead frame of claim 9 comprising:

a tie bar connection; and

the first conductor further comprising a third conductive 25
loop coupled to the tie bar connection and configured for noise cancellation.

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